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Tutorial for Cyber-Physical Systems - Discrete Models

Exercise Sheet 9

Exercise 1: Prefixes and Closure I

4 Points

The goal of this task is to get a better understanding of the relation between the set of finite prefixes of a property and the closure (which is defined using the prefixes).

Let P be any LT property. Prove the following claims:

- (a) $P \subseteq cl(P)$
- (b) $pref(cl(P)) = pref(P)$
- (c) $cl(cl(P)) = cl(P)$

Note: You can use claim (a) in the proof of claim (b), and you can use claims (a) and (b) in the proof of claim (c).

Exercise 2: Prefixes and Closure II

6 Points

The goal of this task is to get a better understanding of prefixes and closures by applying them to given properties.

Consider following properties over the set $AP = \{a, b\}$ of atomic propositions.

- (P_1) a holds exactly once. ↗
- (P_2) Whenever a holds, b holds in the next step.
- (P_3) a holds only finitely many times.
- (P_4) a holds initially and infinitely often. *increases*

For each property P_i complete the following tasks:

- (a) Formalize P_i as a set of traces using set comprehension.
- (b) Give the set of prefixes using set comprehension, i.e. $pref(P_i)$.
- (c) Provide its closure using set comprehension, i.e. $cl(P_i)$.

they are finite.

Exercise 9:

$$cl(P) = \{b \in (2^{APW}) \mid \text{pref}(b) \subseteq \text{pref}(P)\}$$

4/4

a. Take any $b \in P$.

$$\text{pref}(b) \subseteq \text{pref}(P)$$

But since $b \in P$, every finite prefix of P belongs to $\text{pref}(P)$

Thus $\text{pref}(b) \subseteq \text{pref}(P)$ and therefore $b \in cl(P)$

$$\therefore P \subseteq cl(P)$$

✓

b.

$$(i) \text{ pref}(P) \subseteq \text{pref}(cl(P))$$

$$\text{From (a)} \therefore P \subseteq cl(P)$$

$$\therefore \text{pref}(P) \subseteq \text{pref}(cl(P))$$

✓

$$(ii) \text{ pref}(cl(P)) \subseteq \text{pref}(P)$$

Take any prefix $\pi \in \text{pref}(cl(P))$

So there is $b \in cl(P)$ and $\pi \in \text{pref}(b)$

Since $b \in cl(P)$, by definition:

$$\text{pref}(b) \subseteq \text{pref}(P)$$

$$\therefore \pi \in \text{pref}(P)$$

$$\therefore \text{pref}(cl(P)) \subseteq \text{pref}(P)$$

✓

$$(iii) cl(cl(P)) = cl(P)$$

$$\overset{0}{cl(P)} \subseteq cl(cl(P))$$

From a applied to the LT property of $cl(P)$

$$\therefore cl(P) \subseteq cl(cl(P))$$

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$$(iv) cl(cl(P)) \subseteq cl(P)$$

Take any $\epsilon \in cl(cl(P))$

$$\therefore \text{pref}(\epsilon) \subseteq \text{pref}(cl(P))$$

from (b): $\forall \text{pref}(cl(p)) = \text{pref}(p)$

$$\therefore \text{pref}(b) \subseteq \text{pref}(p)$$

which is the condition for $b \in cl(p)$, for all $b \in cl(cl(p))$, there is $b \in cl(p)$

$$\therefore cl(cl(p)) \subseteq cl(p)$$



Exercise 2: 5/6 $cl(P) = \{b \in \Sigma^* \mid \text{pref}(b) \subseteq \text{pref}(P)\}$

P_1 : a holds exactly once. Formal/syntactical errors: -0.5

$$P_1 = \{A_0 A_1 A_2 \dots \in \Sigma^* \mid \exists i \in \mathbb{N}, a \in A_i, \forall j \neq i, a \notin A_j\}$$

$$\text{Pref}(P_1) = \{A_0 \dots A_n \in \Sigma^* \mid \exists i \in \{0, 1, \dots, n\}, a \in A_i\}$$

$$cl(P_1) = \{A_0 A_1 A_2 \dots \in \Sigma^* \mid \exists i \in \mathbb{N}, a \in A_i\}$$

P_2 : whenever a holds, b holds in the next step.

$$P_2 = \{A_0 A_1 A_2 \dots \in \Sigma^* \mid \forall i \in \mathbb{N}, a \in A_i \rightarrow b \in A_{i+1}\}$$

$$\text{Pref}(P_2) = \{A_0 \dots A_n \in \Sigma^* \mid \forall i \in \{0, \dots, n-1\}, a \in A_i \rightarrow b \in A_{i+1}\}$$

$cl(P_2)$: if $b = A_0 A_1 A_2 \dots$
 $\text{pref}(b) \subseteq \text{pref}(P_2) \Leftrightarrow \forall i, a \in A_i \rightarrow b \in A_{i+1}$
 so $cl(P_2) = P_2$.

P_3 : a holds only finitely many times.

$$P_3 = \{A_0 A_1 A_2 \dots \in \Sigma^* \mid \exists j \in \mathbb{N}, \forall i \geq j, a \notin A_i\}$$

$$\text{Pref}(P_3) = \Sigma^+$$

$$cl(P_3) = \Sigma^*$$

P_4 : a holds initially and infinitely often.

$$P_4 = \{A_0 A_1 A_2 \dots \in \Sigma^* \mid a \in A_0 \wedge (\forall n \in \mathbb{N}, \exists i \geq n, a \in A_i)\}$$

$$\text{Pref}(P_4) = \{A_0 \dots A_n \in \Sigma^* \mid a \in A_0\}$$

$$cl(P_4) = \{A_0 A_1 \dots \in \Sigma^* \mid a \in A_0\}$$

Exercise 3: Safety & Liveness Properties

9 Points

The goal of this task is to learn how to recognize invariants, safety and liveness properties, and to learn how one can show that a property belongs to one of these three classes.

In last week's exercise sheet, we already proved and used the following proposition to show that an LT property is not an invariant.

Proposition: Let $E \subseteq (2^{AP})^\omega$ be an LT property. E is not an invariant if and only if there exists a trace $\sigma = A_0A_1 \dots$ such that $\sigma \notin E$, but for every $i \in \mathbb{N}$, the set A_i also occurs in some trace $\sigma_i \in E$.

Consider following properties over the set $AP = \{a, b\}$ of atomic propositions.

- $P_1 = \{ A_0A_1A_2 \dots \in (2^{AP})^\omega \mid \neg \exists i \in \mathbb{N}. a \in A_i \}$
(a never holds)
- $P_2 = \{ A_0A_1A_2 \dots \in (2^{AP})^\omega \mid \forall i \in \mathbb{N}. (a \in A_i \rightarrow \exists j \in \mathbb{N}. (i \leq j \wedge b \in A_j)) \}$
(every a should eventually be followed by b)
- $P_3 = \{ A_0A_1A_2 \dots \in (2^{AP})^\omega \mid \forall i \in \mathbb{N}. (b \in A_i \rightarrow \forall j \in \mathbb{N}. (i \neq j \rightarrow b \notin A_j)) \}$
(b holds at most once)

For each property P_i complete the following tasks:

- (a) Determine if P_i is an invariant. In that case, provide the invariant condition. Otherwise give a trace $\sigma = A_0A_1 \dots$ such that $\sigma \notin P_i$, but for every $i \in \mathbb{N}$, the set A_i also occurs in some trace $\sigma_i \in P_i$.

Example: The property “*in the first step, a holds*” is not an invariant.

We choose $\sigma = (\{b\})^\omega \notin P_{\text{initially } a}$. The set $\{b\}$ also occurs in the trace $\{a\}\{b\}^\omega \in P_{\text{initially } a}$, and similarly $\{b\}$ also occurs in $\{a\}\{b\}^\omega \in P_{\text{initially } a}$.

- (b) Determine if P_i is a safety property. In that case, give the set of all bad prefixes. Otherwise give a counterexample, i.e. a trace $\sigma \in (2^{AP})^\omega \setminus P_i$ such that σ does not have a bad prefix.

Example: The property “*always a*” is a safety property, and its bad prefixes are

$$\text{BadPref}_{\text{always } a} = \{ A_0A_1 \dots A_n \in (2^{AP})^+ \mid \exists i \in \{0, \dots, n\}. a \notin A_i \}$$

- (c) Determine if P_i is a liveness property. In that case, show how any prefix $A_0A_1 \dots A_n$ can be extended to an infinite trace that satisfies P_i . Otherwise give one bad prefix of the property.

Example: The property “*a holds infinitely often*” is a liveness property, and any finite trace prefix $A_0A_1 \dots A_n$ can be extended to an infinite trace σ that satisfies the property by setting $\sigma = A_0A_1 \dots A_n \{a\}^\omega$.

Task 3:

8.5/9

P_1 : a never holds.

Invariant

when $\Phi = \neg a$

$$a \in P_1 \Leftrightarrow \forall i \in \mathbb{N}_0, A_i \models \neg a$$

So P_1 is invariant and invariant condition is $\neg a$

Safety: Yes. A bad prefix is any sequence containing at least one a once it appears, the property is violated.

So P_1 is safety. $\text{Bad-pref}_{P_1} = \{a_0 \dots a_n \in \Sigma^+ \mid \exists i \in \{0 \dots n\}, a \in A_i\}$

Liveness: No. The prefix $\{a\}$ cannot be extended to a word without no occurrence of a

P_2 : (every a followed by b)

invariant: No.

$G = \{a\} \phi^W \dots$ a appears in A_i and there is no b later.

For $A_0 = \{a\}$, $G_0 = \{a\} \{b\}^W \dots$, $G_0 \in P_2$

For $A_i = \phi^W \dots$ ($i \geq 1$), $G_1 = \phi^W \in P_2$

G violates P_2 , but for each position i , we can find a trace in P_2 containing A_i

Safety: No. For any finite prefix, even if it occurs without b afterward, the violation is not yet certain, and b can always appear later.
Counter example! Give a concrete trace without bad prefixes. - 0.5

Liveness: Yes. Every finite prefix can be extended to satisfy eventually b requirement.
Having finite prefix $w = a_0 \dots a_n$: $G = a_0 \dots a_n \{b\} \phi^W \dots$
 $\hookrightarrow b$ can appear later

P_3 (b holds at most once)

Invariant: No.

For trace: $\{b\}\{b\}\{\emptyset\}^W$.

the occurrence of b is twice, so $2 \notin P_2$.

And $A_0 = \{b\}$, $A_1 = \{b\}$, $A_i = \emptyset (i \geq 2)$

For A_0, A_1 , when $b_0 = \{b\}\{\emptyset\}^W$, the appearance of b is one $\rightarrow b_0 \in P_2$.

For A_i , when $b_i = \emptyset^W$, and now there is no b , $b_i \in P_2$. ✓

so P is not invariant.

Safety: ✓ $\{es\}$ ✓

If we want to violate P_3 , the finite prefix must contain two occurrences of b .

$\text{Bad prefix}_3 = \{A_0..A_n \mid \exists i < j \leq n. b \in A_i \wedge b \in A_j\}$ ✓

Liveness: No. ✓ For prefix $W = \{b\}\{b\}$ already violates "at most once" and can not be extension. ✓